

APPROXIMATE FORMS FOR SU(2) MULTI-INSTANTON DETERMINANTS

H. OSBORN and G.P. MOODY

Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Silver Street, Cambridge CB3 9EW, UK

Received 8 April 1980

An approximate form for SU(2) multi-instanton determinants, in both the 2- and 3-dimensional representations, is obtained which incorporates the correct properties under conformal transformations and also the leading singular behaviour when the multi-instanton solution becomes degenerate, at least for the 't Hooft multi-instanton solution.

One approach to understanding non-perturbative effects in quantized gauge theories is hopefully to systematically approximate the functional integrals over the equivalence classes of gauge fields under gauge transformations. This function space is very complicated but may perhaps be represented as far as its main topological features by the stationary points of the euclidean Yang–Mills action. The only known exact stationary points are the self-dual multi-instanton solutions A_μ that are strict minima of the action which is then proportional to the topological index, or instanton number, k . The fluctuations around such a stationary point A_μ involve functional determinants such as $\det(-D^2)$ where the covariant derivative $D_\mu = \partial_\mu + A_\mu$ (determinants for differing spin operators can be trivially related to $\det(-D^2)$, at least for flat space). Expressions for $\det(-D^2)$, where D_μ corresponded to the 2-dimensional representation of SU(2) and A_μ was given by 't Hooft's multi-instanton solution were first obtained by Brown and Creamer [1] in the form of a four-dimensional integral (see also [2]). More recently, integral expressions have been found [3, 4] corresponding to the general self-dual multi-instanton solution for A_μ of ADHM [5] and these results have subsequently been extended to tensor products [6] so that $\det(-D^2)$ can now be given to the same degree of explicitness for D_μ in the adjoint as well as the fundamental representation of an arbitrary gauge group. However, if the functional integral is to be reduced to finite-dimensional integrals over instanton parameters, and the statistical mechanics of dense multi-instanton configurations discussed, then forms of the determinants in which the dependence on the instanton parameters is manifest are probably necessary. This task is here attempted for the SU(2) case.

The initial result [3, 4] for the determinant for D_μ in the fundamental representation with A_μ given by a general self-dual ADHM solution can be written as [4]

$$\begin{aligned}\mathcal{D}_k &= -\ln \{ \det (-D^2/\mu^2) / \det (-D_0^2/\mu^2) \} \\ &= \frac{1}{12}(I + \theta - k \ln \mu^2) - (\alpha(\tfrac{1}{2}) + \tfrac{7}{36})k,\end{aligned}\quad (1)$$

where $D_{0\mu}$ is the trivial covariant derivative for $k=0$ and μ a regularization mass scale. The ratio of determinants in (1) is then finite on flat space and with zeta-function regularization $\alpha(\frac{1}{2}) = -2\zeta'(-1) - \frac{1}{6} \ln 2 - \frac{5}{72}$. I and θ are both integrals over matrices that arise in the ADHM construction and thus depend on the instanton parameters. θ is not expressible directly as a 4-dimensional integral but is a conformal invariant and appears similar in character to the integrals defining topological invariants. I is more straightforwardly given in terms of the fundamental matrix $f(x)$ which acts on a k -dimensional space W and which incorporates the basic reality constraints of the ADHM construction,

$$\begin{aligned}I &= \frac{1}{\pi^2} \int_{\text{reg}} d^4x \{ 5 \operatorname{tr} (f\nu f\nu) - \tfrac{1}{4} \operatorname{tr} (k^2 k^2) \}, \\ f(x)^{-1} &\sim \nu |x|^2, \quad x \rightarrow \infty, \quad k_\alpha = f^{-1} \partial_\alpha f.\end{aligned}\quad (2)$$

For such logarithmically divergent integrals for large x the regularization device adopted is that

$$\frac{1}{\pi^2} \int_{\text{reg}} d^4x \mathcal{J} = \lim_{R^2 \rightarrow \infty} \left\{ \frac{1}{\pi^2} \int_{|x|^2 < R^2} d^4x \mathcal{J} - C \ln R^2 \right\}, \quad \mathcal{J} \underset{|x| \rightarrow \infty}{\sim} \frac{C}{|x|^4}.$$

For the case of $SU(2)$ the results of Jack [6] imply that for the 3-dimensional adjoint representation (denoting corresponding quantities by a tilde) that*

$$\tilde{\mathcal{D}}_k = 4\mathcal{D}_k + \ln \det \{M(\nu \otimes \nu)\} + J, \quad (3)$$

and also

$$\mathcal{D}_k = \frac{1}{6}(\ln \det \{M_S(\nu \otimes \nu)\} + k \ln 2) + \frac{1}{12}J - (\alpha(\tfrac{1}{2}) + \tfrac{5}{72})k - \tfrac{1}{12}k \ln \mu^2. \quad (4)$$

Here J is the finite integral

$$J = -\frac{1}{16\pi^2} \int d^4x \ln \det f\nu \partial^2 \partial^2 \ln \det f\nu, \quad (5)$$

and M is a matrix arising in the extension of the ADHM construction to tensor products, see refs. [6, 7] for definitions, acting on the space $W \times W$; M_S, M_A are the

* We evaluate the undetermined constants in ref. [6] by going to the case of $\operatorname{Sp}(k)$ or $\operatorname{Sp}(k) \times \operatorname{Sp}(k)$ and considering k commuting $\operatorname{Sp}(1)$ factors when the eigenvalues of $-D^2$ or $-\tilde{D}^2$ can be determined, as in ref. [3], but here in the context of zeta-function regularization.

restrictions to the $\frac{1}{2}k(k+1)$, $\frac{1}{2}k(k-1)$ dimensional symmetric, antisymmetric subspaces of $W \times W$, and $\det M = \det M_S \det M_A$. The results (1) and (4) correspond, for $SU(2)$, to

$$I + \theta = J + 2 \ln \det \{M_S(\nu \otimes \nu)\} + (2 \ln 2 + \frac{3}{2})k. \quad (6)$$

Apart from such general integral representations the form of the determinant is also constrained by its properties under conformal transformation [7]. Under the action of a conformal change on x , expressed in quaternionic form as

$$x \rightarrow x' = (\alpha x + \beta)(\gamma x + \chi)^{-1}$$

there is a corresponding change in the gauge field $A_\mu \rightarrow A'_\mu$ given by

$$A_\mu(x') dx'_\mu = A'_\mu(x) dx_\mu.$$

The result of this induced conformal change on the instanton parameters, Δ_c , is for the determinant

$$\Delta_c \mathcal{D}_k = \frac{1}{6}(\Delta_c \ln \det \nu - k \ln \kappa), \quad \Delta_c \tilde{\mathcal{D}}_k = 4 \Delta_c \mathcal{D}_k, \quad (7)$$

for

$$\kappa = \det \begin{pmatrix} \alpha & \beta \\ \gamma & \chi \end{pmatrix}.$$

For the explicit 't Hooft solution the matrix $f(x)$ can be obtained in closed form [7, 4] in terms of its parameters as expressed in terms of the basic scalar field

$$\phi_k = \sum_0^k \frac{\lambda_n^2}{|x_n|^2}, \quad x_n = x - y_n \quad (8)$$

(we use the conformally extended form [8]). In this case $\theta = 0$ and I becomes [4]*

$$I[\phi_k] = \frac{1}{\pi^2} \int_{\text{reg}} d^4 x \left\{ \sum_0^k \frac{1}{|x_n|^4} - \frac{1}{16} \partial^2 \ln \phi_k \partial^2 \ln \phi_k \right\} + \frac{1}{2}k, \quad (9)$$

where the first term in the integrand removes singularities at $x_n = 0$ from the second term. By manipulation of (9) it is possible to directly verify (6) [9, 4] (using $\partial^2 \partial^2 \ln |x_n|^2 = -16 \pi^2 \delta^4(x_n)$) since

$$\ln \det f\nu = -\ln \phi_k P + \ln \sum_0^k \lambda_n^2, \quad P = \prod_0^k |x_n|^2,$$

and [8]

$$\ln \det \{M_S(\nu \otimes \nu)\} + k \ln 2 = (k+1) \left(\ln \sum_0^k \lambda_n^2 - \ln \prod_0^k \lambda_n^2 \right) + \sum_{m \leq n}^k \ln t_{mn}, \quad (10)$$

for $t_{mn} = \lambda_m^2 \lambda_n^2 / |y_m - y_n|^2$, $m \neq n$, $t_{nn} = 0$. Also, in terms of the 't Hooft solution

* In eq. (3.10) of ref. [4] the terms involving $\pi^2 \delta^4(x_i)$ should have an extra factor of $\frac{1}{2}$.

parameters λ_m^2, y_n , we have

$$\ln \det \{M_A(\nu \otimes \nu)\} = (k-1) \left(\ln \sum_0^k \lambda_n^2 - \ln \prod_0^k \lambda_n^2 \right) + \sum_{\substack{m < n \\ 0}}^k \ln t_{mn} - \ln \det p_k, \quad (11)$$

for p_k a $k \times k$ matrix,

$$(p_k)_{nm} = \sum_{l=0}^k t_{nl} \delta_{lm} - t_{nm}, \quad (12)$$

whose determinant can be expanded in products of t 's constrained by rules given in [8]. M_S and M_A are conformal invariants up to a factor so that

$$\Delta_c \ln \det M_S = -k(k+1) \ln \kappa, \quad \Delta_c \ln \det M_A = -k(k-1) \ln \kappa. \quad (13)$$

To analyse the determinants further we investigate the singularities as a function of the instanton parameters. These occur where the instanton configuration degenerates to one corresponding to a lower topological index. For the 't Hooft case this occurs when $y_i \rightarrow y_j$ or $\lambda_i^2 \rightarrow 0$ (or $|y_i|^2 \rightarrow \infty$). To analyse the first limit we let

$$\phi_1^{(ij)} = \frac{\lambda_i^2}{|x_i|^2} + \frac{\lambda_j^2}{|x_j|^2}, \quad (14)$$

and then (9) can be rewritten for this purpose as

$$\begin{aligned} I[\phi_k] = & \frac{1}{\pi^2} \int_{\text{reg}} d^4x \left\{ \sum_{n \neq i,j} \frac{1}{|x_n|^4} + \frac{1}{16} \partial^2 \ln \phi_1^{(ij)} \partial^n \ln \phi_1^{(ij)} - \frac{1}{16} \partial^2 \ln \phi_k \partial^2 \ln \phi_k \right\} \\ & + I[\phi_1^{(ij)}] + \frac{1}{2}(k-1). \end{aligned} \quad (15)$$

$I[\phi_1^{(ij)}]$ can be directly evaluated:

$$I[\phi_1^{(ij)}] = -\ln \frac{\lambda_i^2 \lambda_j^2 |y_i - y_j|^2}{(\lambda_i^2 + \lambda_j^2)^2} + \frac{7}{3},$$

while careful inspection of the integrand in (15) shows that the potential divergence as $y_i \rightarrow y_j$ has been removed and the limit can be taken inside the integral with

$$\frac{1}{4} \partial^2 \phi_1^{(ij)} \rightarrow \frac{1}{|x_i|^2}.$$

Thus we find

$$I[\phi_k] \underset{y_i \rightarrow y_j}{\sim} -\ln \frac{\lambda_i^2 \lambda_j^2 |y_i - y_j|^2}{(\lambda_i^2 + \lambda_j^2)^2} + \frac{7}{3} + I[\bar{\phi}_{k-1}], \quad (16)$$

where $\bar{\phi}_{k-1}$ is the limit of ϕ_k as $y_i \rightarrow y_j$:

$$\bar{\phi}_{k-1} = \sum_{\substack{n \\ n \neq j}}^k \frac{\bar{\lambda}_n^2}{|x_n|^2}, \quad \bar{\lambda}_n^2 = \lambda_n^2, \quad n \neq i, \quad \bar{\lambda}_i^2 = \lambda_i^2 + \lambda_j^2. \quad (17)$$

Now considering $\lambda_i^2 \rightarrow 0$, the resulting divergence arises from terms in the integrand of (9) of order $1/|x_i|^4$ as $x_i \rightarrow 0$. When these are isolated explicitly, then in the remainder $\lambda_i^2 \rightarrow 0$ without problem. Hence we obtain

$$I[\phi_k] \underset{\lambda_i^2 \rightarrow 0}{\sim} \frac{1}{\pi^2} \int_{\text{reg}} d^4x \left\{ \frac{1}{|x_i|^4} - \frac{1}{\phi_k^4} \frac{\lambda_i^8}{|x_i|^{12}} \right\} + \frac{1}{2} + I[\phi'_{k-1}], \quad (18)$$

where

$$\phi'_{k-1} = \sum_{\substack{n=0 \\ n \neq i}}^k \frac{\lambda_n^2}{|x_n|^2}.$$

Setting $u = x_i/\lambda_i$ in (18) so that

$$\phi_k(x) \sim \frac{1}{|u|^2} + \phi'_{k-1}(y_i) + O(\lambda_i^2),$$

allows the integral to be evaluated as $\lambda_i^2 \rightarrow 0$ giving

$$I[\phi_k] \underset{\lambda_i^2 \rightarrow 0}{\sim} -\ln \frac{\lambda_i^2}{\phi'_{k-1}(y_i)} + \frac{7}{3} + I[\phi'_{k-1}]. \quad (19)$$

The right-hand side of (19) is also correct in the essentially equivalent limit $|y_i|^2 \rightarrow \infty$.

We now exhibit an ansatz for $I[\phi_k]$ in the 't Hooft case which satisfies directly the limiting relations (16) and (19):

$$I^0 = 2 \left(\ln \sum_0^k \lambda_n^2 - \ln \prod_0^k \lambda_n^2 \right) + \ln \det p_k + \frac{7}{3}k, \quad (20)$$

or, using (10) and (11), we can naturally extend (20) to the complete SU(2) ADHM solution by defining

$$I^0 + \theta^0 = \ln \det \{M_S(\nu \otimes \nu)\} - \ln \det \{M_A(\nu \otimes \nu)\} + (\ln 2 + \frac{7}{3})k, \quad (21)$$

which checks by (1), (7) and (13) that (20) and (21) have the same properties under conformal transformations as do I or $I + \theta$. Verification that (20) satisfies (16) and (19) depends on the detailed properties of $\det p_k$. Thus

$$\det p_k \underset{t_{ij} \rightarrow \infty}{\sim} t_{ij} \det \bar{p}_{k-1} \quad (22)$$

where $\bar{p}_{k-1}$ is a $(k-1) \times (k-1)$ matrix formed from p_k by eliminating the j th row and column and with $t_{i,n}, t_{n,i} \rightarrow \bar{t}_{i,n} = t_{i,n} + t_{j,n}$. Given the definition of t_{mn} , $\bar{p}_{k-1}$ corresponds exactly to $\bar{\phi}_{k-1}$ in the same way as p_k corresponds to ϕ_k . Furthermore

$$\det p_k \underset{t_{in} \rightarrow 0}{\sim} \sum_{n=0}^k t_{in} \det p'_{k-1}, \quad (23)$$

where p'_{k-1} is a $(k-1) \times (k-1)$ matrix constructed as in (12) from $t_{n,m}$, $n, m \neq i$. These results (22) and (23) may readily be checked by using the complete symmetry of $\det p_k$ under permutations of the $t_{n,m}$ and considering the special cases $t_{01} \rightarrow \infty$ or $t_{0n} \rightarrow 0$. Since $t_{in} \rightarrow 0$ if either $\lambda_i^2 \rightarrow 0$ or $|y_i|^2 \rightarrow \infty$ with

$$\sum_{n=0}^k t_{in} = \lambda_i^2 \phi'_{k-1}(y_i)$$

and p'_{k-1} corresponds appropriately to ϕ'_{k-1} , (23) ensures that (20) is compatible with (18) and (22) also reproduces (16). Following from (21) there is a corresponding form J^0 which reproduces the leading singular behaviour and conformal properties of J :

$$J^0 = -\ln \det \{M(\nu \otimes \nu)\} + (\frac{5}{6} - \ln 2)k. \quad (24)$$

In view of their construction the results (20), (21) and (24) should perhaps be good approximations to the corresponding complete expressions that appear in the $SU(2)$ multi-instanton determinants. We have checked $I[\phi_k] \approx I^0$ by evaluating $I[\phi_k]$ numerically on a computer for $k=2$ and $k=3$ with collinear instantons and in the restricted 't Hooft case where $\lambda_0^2, |y_0|^2 \rightarrow \infty$ with $\lambda_0^2/|y_0|^2 \rightarrow 1$. The agreement is remarkably good, to within about 0.1% for a wide range of parameters, except near singular points when the numerical evaluation fails. Details will be given elsewhere. For $k=2$ in the restricted 't Hooft case we have then

$$\mathcal{D}_2 \approx -\frac{1}{12}(\ln \mu^2 \lambda_1^2 + \ln \mu^2 \lambda_2^2) - 2\alpha(\frac{1}{2}) + \frac{1}{12} \ln \frac{\lambda_1^2 + \lambda_2^2 + |y_1 - y_2|^2}{|y_1 - y_2|^2}, \quad (25)$$

where the last term in (25) is then the correction to the corresponding dilute-gas expressions given by supposing that $\det(-D^2)$ factorises into a product of single instanton determinants. It interpolates smoothly between the asymptotic forms given by Brown and Creamer [1]. More generally we should expect

$$I[\phi_2] = I^0_{k=2} + f(c) \quad (26)$$

for some function of the unique conformal invariant c for $k=2$ and $f(0)=0$, taking $c=0$ in the limits $\lambda_i^2 \rightarrow 0$ or $y_i \rightarrow y_j$. From (13) and (10), (11) [7]

$$c = \frac{t_{01}^2 t_{02}^2 t_{12}^2}{(t_{01} t_{02} + t_{01} t_{12} + t_{02} t_{12})^3} \rightarrow \frac{\lambda_1^2 \lambda_2^2 |y_1 - y_2|^2}{(\lambda_1^2 + \lambda_2^2 + |y_1 - y_2|^2)^3}, \quad (27)$$

where the second form represents the restricted 't Hooft case.

Following the previously outlined procedure and carefully expanding $I[\phi_2]$ to $O(\lambda_1^2)$ it follows that $f'(0)=1^*$. Of course it is then necessary that the difference between $I[\phi_k]$ and I^0 for general k should be a conformal invariant function of the

* If $J \approx J^0$ in (24) then $\tilde{\mathcal{D}}_k \approx 4\mathcal{D}_k + (\frac{5}{6} - \ln 2)k$. For $k=2$ the difference between these is called by Brown and Creamer $-W_2$. Our result that this difference for $k=2$ is $\sim c$ as $c \rightarrow 0$ is compatible with their large separation results, eq. (4.14), though not with the short-distance limit, (4.15).

instanton parameters which reduces in consistent way for degenerate instanton configurations eventually to $f(c)$ for $k = 2$. It should perhaps be noted that for c given by (27), then $c < \frac{1}{27}$ so its effects could well be small. Possibly there may be a systematic expansion of $I + \theta$ or J , for $SU(2)$, with our ansatz as the leading term.

Inspection of the result (25) suggests a long-range coulombic-type $|y_i - y_j|^{-2}$ interaction between well-separated instantons. Bernard [11] has claimed that such effects will be cancelled by contributions due to the jacobian in the transformation of the functional integral to one over instanton parameters. Clearly it would be desirable to investigate this more systematically.

We would like to thank Dr. P. Goddard for many helpful conversations and G.P. Moody would like to thank the SRC for a research grant.

References

- [1] L.S. Brown and D.B. Creamer, Phys. Rev. D18 (1978) 3695
- [2] C. Callias and C.H. Taubes, MIT preprint CTP 783
- [3] B. Berg and M. Luscher, Nucl. Phys. B160 (1979) 281
- [4] H. Osborn, Nucl. Phys. B159 (1979) 497
- [5] M.F. Atiyah, N.J. Hitchin, V.G. Drinfeld and Yu.I. Manin, Phys. Lett. 65A (1978) 185
- [6] I. Jack, DAMTP preprint 79/21, Nucl. Phys., to be published
- [7] E. Corrigan, P. Goddard, H. Osborn and S. Templeton, Nucl. Phys. B159 (1979) 469
- [8] R. Jackiw, C. Nohl and C. Rebbi, Phys. Rev. D15 (1977) 1642
- [9] A.A. Belavin, V.A. Fateev, A.S. Schwarz and Yu.S. Tyupkin, Phys. Lett. 83B (1979) 317
- [10] C.W. Borchardt, Crelle's Jour. 57 (1859) 111
- [11] C. Bernard, Phys. Rev. D18 (1978) 2026